1.1

$$V = ax^2 + by^2 \quad \longrightarrow \quad L = \frac{m}{2}(\dot{x}^2+\dot{y}^2) - (ax^2+by^2)$$

$$\begin{cases} m\ddot{x} = -2ax \\ m\ddot{y} = -2by \end{cases} \qquad \begin{cases} P_x = \dfrac{\partial L}{\partial \dot{x}} = m\dot{x} \\ P_y = \dfrac{\partial L}{\partial \dot{y}} = m\dot{y} \end{cases}$$

$$H = P_x\dot{x} + P_y\dot{y} - L = \frac{1}{2m}(P_x^2+P_y^2) + (ax^2+by^2)$$

$$\begin{cases} \dot{x} = \dfrac{\partial H}{\partial P_x} = \{x, H\} = \dfrac{P_x}{m} \\ \dot{y} = \dfrac{\partial H}{\partial P_y} = \{y, H\} = \dfrac{P_y}{m} \end{cases} \qquad \begin{cases} \dot{P}_x = -\dfrac{\partial H}{\partial x} = -\{P_x, H\} = -2ax \\ \dot{P}_y = -\dfrac{\partial H}{\partial y} = -\{P_y, H\} = -2by \end{cases}$$

$$E_x = \frac{P_x^2}{2m} + ax^2 \quad \longrightarrow \quad \frac{dE_x}{dt} = 0 \quad : \text{運動の恒量}$$

$$\longrightarrow \quad \frac{x^2}{\frac{E_x}{a}} + \frac{P_x^2}{2mE_x} = 1$$

$$\oint P_x\,dx = \pi\sqrt{\frac{E_x}{a}}\sqrt{2mE_x} = nh + \text{const}$$

$$\therefore \quad E_x = \frac{nh}{\pi}\sqrt{\frac{a}{2m}} + \text{const}$$

$$V = -\frac{e^2}{r} \rightarrow L = \frac{m}{2}(\dot{r}^2 + r^2\dot{\theta}^2) + \frac{e^2}{r}$$

$$\begin{cases} m\ddot{r} = -\dfrac{e^2}{r^2} \\ \dfrac{d}{dt}(mr^2\dot{\theta}) = 0 \end{cases} \qquad \begin{cases} P_r = \dfrac{\partial L}{\partial \dot{r}} = m\dot{r} \\ P_\theta = \dfrac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta} \end{cases}$$

$$H = P_r\dot{r} + P_\theta\dot{\theta} - L = \frac{1}{2m}\left(P_r^2 + \frac{P_\theta^2}{r^2}\right) - \frac{e^2}{r}$$

$$\begin{cases} \dot{r} = \dfrac{\partial H}{\partial P_r} = \{r, H\} = \dfrac{P_r}{m} \\ \dot{\theta} = \dfrac{\partial H}{\partial P_\theta} = \{\theta, H\} = \dfrac{P_\theta}{mr^2} \end{cases} \qquad \begin{cases} \dot{P}_r = -\dfrac{\partial H}{\partial r} = -\{P_r, H\} = \dfrac{P_\theta^2}{mr^3} - \dfrac{e^2}{r^2} \\ \dot{P}_\theta = -\dfrac{\partial H}{\partial \theta} = -\{P_\theta, H\} = 0 \end{cases}$$

$$\frac{dP_\theta}{dt} = 0 \quad , \quad \frac{dH}{dt} = 0$$

$$\oint P_\theta\, d\theta = 2\pi M \quad (\because P_\theta = \text{const}) \qquad M = \frac{nh}{2\pi} + \text{const}$$

$$E = \frac{1}{2m}\left(P_r^2 + \frac{M^2}{r^2}\right) - \frac{e^2}{r} \text{ より}$$

$$\oint P_r\, dr = \oint \sqrt{2mE + \frac{2me^2}{r} - \frac{M^2}{r^2}}\, dr = 2\int_{r_1}^{r_2} \sqrt{-2mE\left(1 - \frac{r_1}{r}\right)\left(\frac{r_2}{r} - 1\right)}\, dr$$

$$= 2\pi\left\{\frac{me^2}{\sqrt{-2mE}} - |M|\right\}$$

$$\therefore E = -\frac{2\pi^2 me^4}{(n+n')^2 h^2}$$

1.2

(1) $$\int_{-\infty}^{\infty} f(x) \left[\frac{\lim_{a\to\infty} a g(ax)}{\int_{-\infty}^{\infty} g(x')\,dx'} \right] dx = \frac{\lim_{\epsilon\to 0} \int_{-\infty}^{\infty} f(\epsilon x)\, g(x)\,dx}{\int_{-\infty}^{\infty} g(x')\,dx'} = f(0)$$

(2) $$\int_{-\infty}^{\infty} g(x)\,\delta(f(x))\,dx = \sum_i \int_{x_i-\epsilon}^{x_i+\epsilon} g(x)\,\delta(f(x))\,dx$$

$$= \sum_i \int_{-\varepsilon}^{\varepsilon} g(x) \left|\frac{df}{dx}\right|^{-1} \delta(f)\,df$$

$$= \sum_i \left|\frac{df}{dx}\right|^{-1}_{x_i} g(x_i)$$

1.3

(1) $\hat{A}_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ は x軸に対する折り返し

$\hat{A}_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\hat{A}_1 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$\hat{A}_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ は $y=x$ に対する折り返し

$\hat{A}_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $\hat{A}_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix} = -\begin{pmatrix} -1 \\ 1 \end{pmatrix}$

(2) 自明

(3) $X = \begin{pmatrix} a \\ b \\ c \end{pmatrix}_{\langle 1, x, x^2 \rangle} \rightarrow \frac{d}{dx} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}_{\langle 1, x, x^2 \rangle}$

(4) 自明 ($\because$ 和と実数倍について閉じているから)

1.4

(1) 略

(2) Jacobi の恒等式

$$[\hat{A}, [\hat{B}, \hat{C}]] + [\hat{B}, [\hat{C}, \hat{A}]] + [\hat{C}, [\hat{A}, \hat{B}]] = 0$$

(3) $[\hat{x}, \hat{p}] = i\hbar$ $\left(\because \frac{d}{dx}x - x\frac{d}{dx} = 1\right)$

2.1

(1) $kx - \omega t = \text{const} \rightarrow v_\phi = \frac{\omega}{k} = \frac{E}{p}$

(2) $v_g = \left(\frac{d\omega}{dk}\right)_{km}$

(3) $v_\phi = c \;,\quad v_\phi = \frac{p}{2m} = \frac{v}{2}$

(4) $i\hbar \frac{\partial \psi}{\partial t} = i\hbar \int dk\, f(k)(-i\omega)\psi_k = E\psi$

$$-\frac{\hbar^2}{2m}\frac{\partial^2 \psi}{\partial x^2} = -\frac{\hbar^2}{2m}\int dk\, f(k)(ik)^2 \psi_k = E\psi$$

$\omega = \omega(k)$

(2) $\psi(x,t) = \frac{1}{\sqrt{2\pi}} \int dk\, f(k) e^{i(kx-\omega t)}$

$$\langle x \rangle = \int \psi^* \hat{x} \psi \, dx = \frac{1}{2\pi} \int dx\, dk\, dk'\, f^*(k)\, \hat{x}\, f(k')\, e^{-i(k-k')x}\, e^{i(\omega-\omega')t}$$

$$= \frac{1}{2\pi} \int dx\, dk\, dk'\, f^*(k) \frac{1}{i}\frac{\partial}{\partial k'}\left(e^{-i(k-k')x}\right) f(k')\, e^{i(\omega-\omega')t}$$

$$= \int dk\, dk'\, f^*(k) \frac{1}{i}\frac{\partial}{\partial k'} \delta(k-k') f(k')\, e^{i(\omega-\omega')t}$$

$$v_g = \frac{\partial \langle x \rangle}{\partial t} = \int dk\, dk'\, f^*(k) \frac{\partial}{\partial k'} \delta(k-k') f(k') (\omega' - \omega) e^{i(\omega-\omega')t}$$

$$= \int dk\, dk'\, f^*(k)\, \delta(k-k')\, f(k') \left(\frac{\partial \omega'}{\partial k'}\right) e^{i(\omega-\omega')t}$$

$$= \int dk\, f^*(k) f(k) \left(\frac{\partial \omega}{\partial k}\right)$$

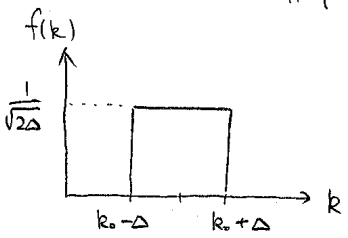
2.2

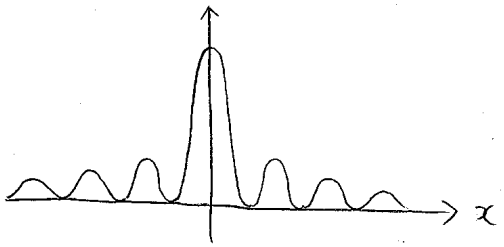
(1) $$\psi(x) = \int_{-\infty}^{\infty} f(k)\,\psi_k(x,0)\,dk = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(k)\,e^{ikx}\,dk$$

$$f(k) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \psi(x)\,e^{-ikx}\,dx$$

(2) $$\psi(x) = \frac{1}{\sqrt{2\pi}}\int_{k_0-\Delta}^{k_0+\Delta} \frac{1}{\sqrt{2\Delta}}\,e^{ikx}\,dk = \sqrt{\frac{\Delta}{\pi}}\,e^{ik_0x}\,\frac{\sin\Delta x}{\Delta x}$$

$$|\psi(x)|^2 = \frac{\Delta}{\pi}\left(\frac{\sin\Delta x}{\Delta x}\right)^2$$

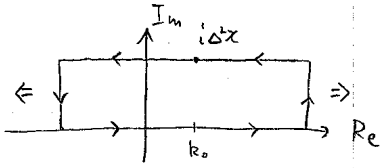




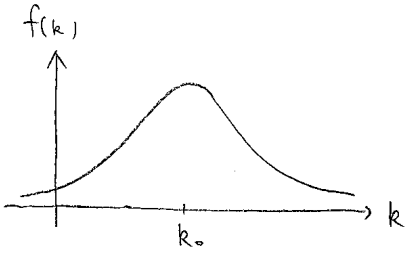
$$\psi(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \frac{1}{(\pi\Delta^2)^{1/4}}\,e^{ikx-\frac{(k-k_0)^2}{2\Delta^2}}\,dk$$

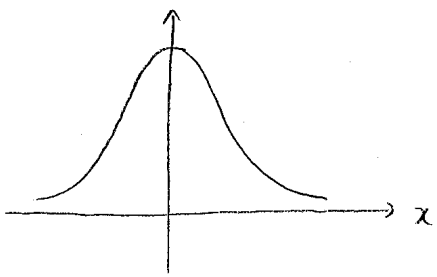
$$= \frac{1}{\sqrt{2\pi}}\cdot\frac{1}{(\pi\Delta^2)^{1/4}}\,e^{ik_0x-\frac{1}{2}\Delta^2x^2}\int_{-\infty}^{\infty} e^{-\frac{\{k-(k_0+i\Delta^2x)\}^2}{2\Delta^2}}\,dk$$

$$= \left(\frac{\Delta^2}{\pi}\right)^{1/4} e^{ik_0x-\frac{1}{2}\Delta^2x^2}$$



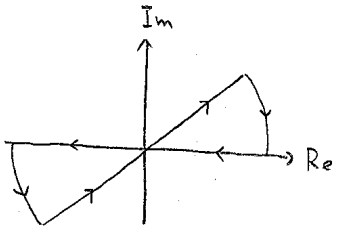
$$|\psi(x)|^2 = \frac{\Delta}{\sqrt{\pi}}\,e^{-\Delta^2x^2}$$





2.3

(1) $-\frac{(k-k_0)^2}{2\Delta^2} + i\left(kx - \frac{\hbar k^2}{2m}t\right)$



$$= -\frac{1}{2}\left(\frac{1}{\Delta^2} + i\frac{\hbar t}{m}\right)\left(k - \frac{\frac{k_0}{\Delta^2} + ix}{\frac{1}{\Delta^2} + i\frac{\hbar t}{m}}\right)^2 - \frac{k_0^2}{2\Delta^2} + \frac{1}{2}\cdot\frac{\left(\frac{k_0}{\Delta^2} + ix\right)^2}{\frac{1}{\Delta^2} + i\frac{\hbar t}{m}}$$

$$\psi(x,t) = \frac{1}{\pi^{1/4}}\left(\frac{\Delta}{1 + i\frac{\hbar t}{m}\Delta^2}\right)^{1/2} e^{i\left(k_0 x - \frac{\hbar k_0^2}{2m}t\right)}\, e^{-\frac{\Delta^2}{2}\cdot\frac{\left(x - \frac{\hbar k_0}{m}t\right)^2}{1 + i\frac{\hbar t}{m}\Delta^2}}$$

$$\langle\hat{x}\rangle = \frac{\hbar k_0}{m}t$$

$$\Delta x = \sqrt{\langle(\hat{x} - \langle\hat{x}\rangle)^2\rangle} = \frac{1}{\sqrt{2}\Delta}\sqrt{1 + \left(\frac{\hbar t}{m}\Delta^2\right)^2}$$

$$\langle\hat{p}\rangle = \hbar k_0$$

$$\Delta p = \sqrt{\langle(\hat{p} - \langle\hat{p}\rangle)^2\rangle} = \frac{\hbar\Delta}{\sqrt{2}} \qquad (t\text{によらない} \rightarrow \text{平面波})$$

(2) $\omega = ck$

$$-\frac{(k-k_0)^2}{2\Delta^2} + ik(x - ct) = -\frac{1}{2\Delta^2}\left(k - k_0 - i\Delta^2(x-ct)\right)^2 - \frac{\Delta^2}{2}(x-ct)^2$$

$$\psi(x,t) = (\pi\Delta^2)^{-1/4}\, e^{-\frac{\Delta^2}{2}(x-ct)^2}$$

$$\langle\hat{x}\rangle = ct$$

2.4

(1) $\langle\varphi|\varphi\rangle = \langle\psi|(\hat{x}-i\lambda\hat{p})(\hat{x}+i\lambda\hat{p})|\psi\rangle = \langle\psi|\hat{x}^2+\lambda^2\hat{p}^2+i\lambda[\hat{x},\hat{p}]|\psi\rangle$

$= \langle\hat{p}^2\rangle\lambda^2 - \hbar\lambda + \langle\hat{x}^2\rangle \geq 0$

$D \leq 0 \rightarrow \langle\hat{x}^2\rangle\langle\hat{p}^2\rangle \geq \frac{\hbar^2}{4}$

(2) $\left.\begin{aligned}\hat{x}' &= \hat{x} - \langle\hat{x}\rangle \\ \hat{p}' &= \hat{p} - \langle\hat{p}\rangle\end{aligned}\right\}$ $[\hat{x}',\hat{p}'] = [\hat{x},\hat{p}] = i\hbar$

$\therefore \langle\hat{x}'^2\rangle\langle\hat{p}'^2\rangle = (\Delta x)^2(\Delta p)^2 \geq \frac{\hbar^2}{4}$

(3) $|\varphi\rangle = (\hat{A}+i\lambda\hat{B})|\psi\rangle$ とおくと

$\langle\hat{B}^2\rangle\lambda^2 + i[\hat{A},\hat{B}]\lambda + \langle\hat{A}^2\rangle \geq 0$

$\rightarrow \langle\hat{A}^2\rangle\langle\hat{B}^2\rangle \geq \frac{1}{4}|[\hat{A},\hat{B}]|^2$

(4) $\langle\varphi|\varphi\rangle = (\Delta p)^2\lambda^2 - \hbar\lambda + (\Delta x)^2 = (\Delta p)^2\left(\lambda - \frac{\hbar}{2(\Delta p)^2}\right)^2 - \frac{\hbar^2}{4(\Delta p)^2} + (\Delta x)^2$

$\left(\hat{x}' + i\frac{2}{\hbar}(\Delta x)^2\hat{p}'\right)|\psi\rangle = 0$

$\lambda = \frac{\hbar}{2(\Delta p)^2} = \frac{2}{\hbar}(\Delta x)^2$

$\hat{x} - \langle\hat{x}\rangle + i\frac{2}{\hbar}(\Delta x)^2(\hat{p} - \langle p\rangle)$

$x\psi + 2(\Delta x)^2\frac{d\psi}{dx} = \left(\langle x\rangle + i\frac{2}{\hbar}(\Delta x)^2\right)\psi$

$\psi(x) = \frac{1}{(2\pi)^{\frac{1}{4}}(\Delta x)^{\frac{1}{2}}}\, e^{-\frac{\left(x - \langle x\rangle - i\frac{\hbar}{2}(\Delta x)^2\right)^2}{4(\Delta x)^2}}$

Gauss 波束

2.5

(1) $G(x', t'; x, t) = \sum_n \langle x' | n \rangle \langle n | x \rangle e^{-\frac{i}{\hbar} E_n (t'-t)}$

$= \langle x' | e^{-\frac{i}{\hbar}\hat{H}(t'-t)} | x \rangle$

$= \int dk \, \langle x' | e^{-\frac{i}{\hbar}\hat{H}(t'-t)} | k \rangle \langle k | x \rangle$

自由粒子

$= \int dk \, e^{-\frac{i}{\hbar}\cdot\left(\frac{\hbar^2 k^2}{2m}\right)(t'-t)} \cdot \frac{1}{\sqrt{2\pi}} e^{ikx'} \cdot \frac{1}{\sqrt{2\pi}} e^{-ikx}$

$= \sqrt{\frac{m}{2\pi i \hbar (t'-t)}} \; e^{\frac{i}{\hbar}\cdot\frac{m}{2}\cdot\frac{(x'-x)^2}{t'-t}}$

$\xrightarrow{t' \to t}$ $\delta(x'-x) = \frac{1}{2\pi}\int dk \, e^{ik(x-x')}$

(2) $\int dx \, G \langle x | \psi \rangle = \int d\xi \, dx \, \langle x' | e^{-\frac{i}{\hbar}\hat{H}(t'-t)} | \xi \rangle \langle \xi | x \rangle \langle x | \psi \rangle$

$= \int d\xi \, e^{-\frac{i}{\hbar} E_\xi (t'-t)} \langle x' | \xi \rangle \langle \xi | \psi \rangle$

$= \langle x' | \psi \rangle$?

(3) $\psi(x', t') = \int dx \, G \langle x | \psi \rangle$ $\qquad \alpha = \frac{\hbar t'}{im}$

$= \int dx \, \frac{1}{\sqrt{2\pi\alpha}} e^{\frac{(x'-x)^2}{2\alpha}} \left|\frac{\Delta^2}{\pi}\right|^{1/4} e^{ik_0 x - \frac{1}{2}\Delta^2 x^2}$

$-\frac{1}{2}\left(\Delta^2 - \frac{1}{\alpha}\right)\left(x - \frac{x' - ik_0\alpha}{1-\alpha\Delta^2}\right)^2 + \frac{1}{2}\frac{\left(x' - \frac{ik_0}{\Delta^2}\right)^2}{\left(\alpha - \frac{1}{\Delta^2}\right)} = \frac{k_0^2}{2\Delta^2}$

$=$

2.6

(1) $\hat{A}\psi = a\psi$

$\langle \hat{A}\psi | \psi \rangle = \langle \psi | \hat{A}^\dagger \psi \rangle \rightarrow a^* \langle \psi | \psi \rangle = a \langle \psi | \psi \rangle$

(2) $\hat{p}$ がエルミートであることを示す。

$$\int (\hat{p}\psi)^* \phi \, dx = \int \left(\frac{\hbar}{i}\frac{\partial \psi}{\partial x}\right)^* \phi \, dx$$

$$= \int -\frac{\hbar}{i}\frac{\partial \psi^*}{\partial x} \phi \, dx$$

$$= \left[-\frac{\hbar}{i}\psi^*\phi\right] + \int \psi^* \frac{\hbar}{i}\frac{\partial \phi}{\partial x} dx$$

$$= \int \psi^* (\hat{p}\phi) dx \quad \lrcorner$$

よって $H = \frac{\hat{p}^2}{2m} + V(\hat{x})$ もエルミート

(3) $i\hbar \frac{\partial \psi}{\partial t} = H\psi$

$-i\hbar \frac{\partial \psi^*}{\partial t} = \psi^* H^\dagger = H\psi^*$

$$\frac{d}{dt}\int \psi^* \psi \, dx = \int \left(\frac{\partial \psi^*}{\partial t}\psi + \psi^* \frac{\partial \psi}{\partial t}\right) dx$$

$$= \frac{1}{i\hbar}\int \left(-(H\psi^*)\psi + \psi^*(H\psi)\right) dx$$

$$= 0$$

3.1

(1) $\hat{P} u_p = p' u_p \qquad u_p = \frac{1}{\sqrt{2\pi\hbar}} e^{\frac{i}{\hbar} p' x}$

$\hat{x} u_x = x' u_x \qquad u_x = \delta(x - x')$

(2) $\psi(x) = \int dp\, f(p) u_p(x)$

$u_x(x) = \delta(x - \xi) = \frac{1}{2\pi\hbar} \int dp' e^{\frac{i}{\hbar} p'(x - x')} \qquad f(p) = \frac{1}{\sqrt{2\pi\hbar}} e^{-\frac{i}{\hbar} p' x'}$

(3) $\int dx\, \psi^*(x)\, \hat{p}\, \varphi(x) = \int dx\, dp\, dp'\, f^*(p)\, u_p^*\, p'\, g(p')\, u_{p'}$

$= \int dx\, dp\, dp'\, f^*(p)\, p'\, g(p')\, \frac{1}{2\pi\hbar} e^{-\frac{i}{\hbar}(p - p')x}$

$= \int dp\, dp'\, f^*(p)\, p'\, g(p')\, \delta(p - p') = \int dp\, f^*(p)\, p\, g(p)$

$\int dx\, \psi^*(x) \hat{x} \varphi(x) = \int dx\, dp\, dp'\, f^*(p)\, u_p^*\, \hat{x}\, g(p')\, u_{p'}$

$= \int dx\, dp\, dp'\, f^*(p)\, \hat{x}'\, g(p')\, \frac{1}{2\pi\hbar} e^{-\frac{i}{\hbar}(p - p')x}$

$= \int dx\, dp\, dp'\, f^*(p)\, \frac{1}{2\pi\hbar} \frac{\hbar}{i} \frac{\partial}{\partial p'} \left(e^{-\frac{i}{\hbar}(p - p')x} \right) g(p')$

$= \int dp\, dp'\, f^*(p)\, \frac{\hbar}{i} \frac{\partial}{\partial p'}\, \delta(p - p')\, g(p') = \int dp\, f^*(p)\, \frac{\hbar}{i} \frac{\partial}{\partial p}\, g(p)$

(4) $\int_{-\infty}^{\infty} \psi^*(x) F(\hat{x}) \varphi(x)\, dx = \int_{-\infty}^{\infty} f^*(p) F\left(i\hbar \frac{d}{dp}\right) g(p)\, dp$

$\int_{-\infty}^{\infty} f^*(p) F(\hat{p}) g(p)\, dp = \int_{-\infty}^{\infty} \psi^*(x) F\left(\frac{\hbar}{i} \frac{d}{dx}\right) \varphi(x)\, dx$

3.2

(1) $\frac{dW}{dx} = \psi_1\psi_2'' - \psi_1''\psi_2 = (V-\epsilon)(\psi_1\psi_2 - \psi_1\psi_2) = 0$ $\epsilon_1 = \epsilon_2$

(2) 縮退しているとすると $W = \text{const.} = 0$ ($|x| \to 0$ で $\psi \to 0$)

$\frac{\psi_1}{\psi_2} = \text{const}$ より 矛盾

(3) $W' = (\epsilon_1 - \epsilon_2)\psi_1\psi_2$ 恒等的に
ゼロではない.

$\epsilon_1 < \epsilon_2$



3.3

$x \to -\infty$	$x \to +\infty$
$\psi_1 = e^{ikx} + Re^{-ikx}$	Te^{ikx}
$\psi_2 = T'e^{+ikx}$	$Re^{ikx} + e^{+ikx}$

$\epsilon_1 = \epsilon_2$ 라 하면 $W(-\infty) = W(+\infty)$

$\alpha = e^{ikx \to +\infty}$

$$-\left(\frac{1}{\alpha} + R\alpha\right)T'\alpha - \left(\frac{1}{\alpha} - R\alpha\right)T'\alpha = T\alpha\left(R'\alpha - \frac{1}{\alpha}\right) - T\alpha\left(R'\alpha + \frac{1}{\alpha}\right)$$

$$T' = T$$

3.4

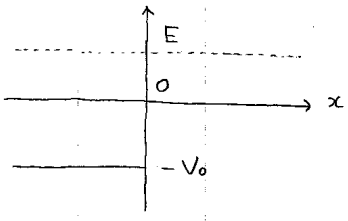
(1)

(2) $\psi_E(x) \to \psi_{E+\Delta E}(x+\Delta x) = \psi_E(x) + \frac{\partial \psi_E}{\partial x}\Delta x + \frac{\partial \psi_E}{\partial E}\Delta E$

(4) $E = \langle\psi|H|\psi\rangle = \frac{1}{2m}\langle\psi|p^2|\psi\rangle + \langle\psi|V|\psi\rangle$

$$= \frac{1}{2m}\underbrace{\langle p^2\rangle}_{(\Delta p)^2 > 0} - V_0 > -V_0$$

3.5



(1) $-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi = E\psi$

$e^{ikx} + \tilde{R}e^{-ikx} \quad \vdots \quad \tilde{T}e^{ik_0x}$

$k = \sqrt{\frac{2m}{\hbar^2}(E+V_0)} \qquad k_0 = \sqrt{\frac{2mE}{\hbar^2}}$

$x=0$; $\quad 1+\tilde{R} = \tilde{T} \qquad T = \tilde{T}^2 = \left(\frac{2}{1+\sqrt{\frac{E}{E+V_0}}}\right)^2$

$k(1-\tilde{R}) = k_0\tilde{T}$

$$R = \tilde{R}^2 = \left(\frac{1-\sqrt{\frac{E}{E+V_0}}}{1+\sqrt{\frac{E}{E+V_0}}}\right)^2$$

(2) $E < 0$

$\tilde{T}e^{-\frac{\sqrt{2m|E|}}{\hbar}x} \qquad \frac{R}{T} = \left|\frac{\tilde{R}}{\tilde{T}}\right|^2 = \frac{1}{4}\left|1-i\sqrt{\frac{-E}{E+V_0}}\right|^2 = \frac{1}{4}\cdot\frac{V_0}{E+V_0}$

$(x>0)$

(3) $R + \frac{k_0}{k}T = 1$

3.6

$\langle x|t=t_0\rangle = \delta(x-x_0) = \frac{1}{2\pi}\int e^{ik(x-x_0)}dk \qquad E_k = \frac{\hbar^2k^2}{2m} - V_0$

$\langle x|\psi\rangle = \langle x|e^{-\frac{i}{\hbar}H(t-t_0)}|t=t_0\rangle = \int dk\,\langle x|e^{-\frac{i}{\hbar}H(t-t_0)}|k\rangle\langle k|t=t_0\rangle$

$= \frac{1}{2\pi}\int dk\, e^{-\frac{i}{\hbar}E_k(t-t_0)}e^{ik(x-x_0)} \qquad E_k = \begin{cases}\frac{\hbar^2k^2}{2m} - V_0 & x<0\\ \frac{\hbar^2k^2}{2m} & x>0\end{cases}$

$= e^{\frac{i}{\hbar}V_0\theta(-x)(t-t_0)}\frac{1}{2\pi}\int dk\, e^{-\frac{i\hbar\tilde{t}}{2m}\left(k-\frac{\tilde{x}}{\hbar\tilde{t}}\right)^2 + i\frac{\tilde{x}^2}{2m\tilde{t}}}$

$\tilde{x} = x - x_0$

$\tilde{t} = t - t_0$

$= e^{\frac{i}{\hbar}V_0\theta(-x)\tilde{t}}e^{i\frac{m}{2\hbar}\frac{\tilde{x}^2}{\tilde{t}}}\frac{1}{2\pi}\int dk\, e^{-\frac{i\hbar\tilde{t}}{2m}k^2}$

$= \sqrt{\frac{m}{2\pi\hbar\tilde{t}}}\,e^{i\frac{m\tilde{x}^2}{2\hbar\tilde{t}}}e^{\frac{i}{\hbar}V_0\theta(-x)\tilde{t}} = \sqrt{\frac{m}{2\pi\hbar(t-t_0)}}\,e^{i\frac{m(x-x_0)^2}{2\hbar(t-t_0)}}e^{\frac{i}{\hbar}V_0\theta(-x)(t-t_0)}$

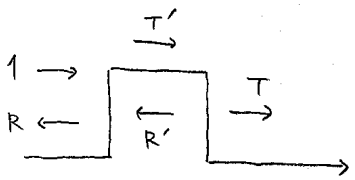
3.7

(1) $e^{ik(-a)} + Re^{-ik(-a)} = T'e^{ik'(-a)} + R'e^{-ik'(-a)}$

$ik\left(e^{ik(-a)} - Re^{-ik(-a)}\right) = ik'\left(T'e^{ik'(-a)} - R'e^{-ik'(-a)}\right)$

$T'e^{ik'a} + R'e^{-ik'a} = Te^{ika}$

$ik'\left(T'e^{ik'a} - R'e^{-ik'a}\right) = ikTe^{ika}$

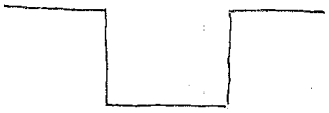


$k = \sqrt{E}$ $\quad k' = \sqrt{E - V_0}$

$\alpha = e^{ika}$ $\quad \alpha' = e^{ik'a}$

$$\rightarrow \begin{cases} \frac{1}{\alpha^2} + R\frac{k'-k}{k'+k} = \frac{T}{\alpha'^2} \\ \frac{1}{\alpha^2} + R\frac{k'+k}{k'-k} = T\alpha'^2 \end{cases}$$

3.8



4.1

$$H = \frac{1}{2m}\hat{P}^2 + \frac{m}{2}\omega^2\hat{x}^2 = \hbar\omega\left(\hat{P}^2 + \hat{Q}^2\right) \qquad \hat{P} = \sqrt{\frac{1}{2m\hbar\omega}}\,\hat{P}\,,\quad \hat{Q} = \sqrt{\frac{m\omega}{2\hbar}}\,\hat{x}$$

(1) $$\begin{cases} \hat{a} = \hat{Q} + i\hat{P} \\ \hat{a}^\dagger = \hat{Q} - i\hat{P} \end{cases} \rightarrow \begin{cases} \hat{P} = \frac{1}{2i}(\hat{a} - \hat{a}^\dagger) \\ \hat{Q} = \frac{1}{2}(\hat{a} + \hat{a}^\dagger) \end{cases}$$

$$[\hat{a}, \hat{a}^\dagger] = 2i[\hat{P}, \hat{Q}] = \frac{i}{\hbar}[\hat{P}, \hat{x}] = 1 \qquad \text{☺ } [\hat{x}, \hat{P}] = i\hbar$$

$$\hat{H} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)$$

$$[\hat{a}, \hat{H}] = \hbar\omega[\hat{a}, \hat{a}^\dagger\hat{a}] = \hbar\omega\hat{a}$$

$$[\hat{a}^\dagger, \hat{H}] = \hbar\omega[\hat{a}^\dagger, \hat{a}^\dagger\hat{a}] = -\hbar\omega\hat{a}^\dagger$$

(2) $$\hat{H}\psi_n = E_n\psi_n$$

$$\hat{H}(\hat{a}\psi_n) = (E_n - \hbar\omega)(\hat{a}\psi_n)$$

$$\hat{H}(\hat{a}^\dagger\psi_n) = (E_n + \hbar\omega)(\hat{a}^\dagger\psi_n)$$

(3) $$0 \le \hbar\omega\langle\hat{a}\psi_n|\hat{a}\psi_n\rangle = \hbar\omega\langle n|\hat{a}^\dagger\hat{a}|n\rangle = \langle n|\left(\hat{H} - \frac{1}{2}\hbar\omega\right)|n\rangle$$

$$= E_n - \frac{1}{2}\hbar\omega \qquad \therefore E_n \ge \frac{1}{2}\hbar\omega$$

$$E_0 = \frac{1}{2}\hbar\omega\,,\quad \psi_0(x) = \left(\frac{m\omega}{\hbar\pi}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2}$$

4.2

(1) $\hat{a}^\dagger |m\rangle = \sqrt{n+1}\,|m+1\rangle$ ☺ $\langle m|\hat{a}\hat{a}^\dagger|m\rangle = n+1$

$\therefore\ |m\rangle = \dfrac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle$ $\qquad E_n = \hbar\omega\left(n+\frac{1}{2}\right)$

(2) $\hat{F}(t) = e^{t\hat{s}}\hat{o}\,e^{-t\hat{s}}$ $\qquad \dfrac{d\hat{F}}{dt} = e^{\hat{s}t}[\hat{s},\hat{o}]e^{-\hat{s}t}$

$$\frac{d^n\hat{F}}{dt^n} = e^{\hat{s}t}[\underbrace{\hat{s},[\hat{s},\cdots,[\hat{s}}_{n\text{個}},\hat{o}]\cdots]]e^{-\hat{s}t}$$

(3) $$-\frac{i}{\sqrt{2m\hbar\omega}}e^{\frac{\alpha^2\hat{x}^2}{2}}\hat{p}\,e^{-\frac{\alpha^2\hat{x}^2}{2}} = -i\,e^{\hat{Q}^2}\hat{P}\,e^{-\hat{Q}^2} = -i\,(\hat{P}+i\hat{Q}+\underbrace{\cdots\cdots}_{=0}) = \hat{a}^\dagger$$

$\left(\text{☺}\ [\hat{Q},\hat{P}] = \frac{i}{2}\right)$

$$\psi_n(x) = \frac{1}{\sqrt{n!}}\left(\frac{-i}{\sqrt{2m\hbar\omega}}\right)^n\left(e^{\frac{\alpha^2\hat{x}^2}{2}}\hat{p}\,e^{-\frac{\alpha^2\hat{x}^2}{2}}\right)^n\psi_0(x) \qquad \left(\hat{p} = -i\hbar\frac{d}{dx}\right)$$

$$= \frac{1}{\sqrt{n!}}\left(-\sqrt{\frac{\hbar}{2m\omega}}\right)^n\left(e^{\frac{\alpha^2x^2}{2}}\frac{d^n}{dx^n}e^{-\frac{\alpha^2x^2}{2}}\right)\psi_0(x)$$

$$= \sqrt{\frac{\sqrt{m\omega}}{n!\sqrt{\hbar}\,\pi}}(-1)^n\left(e^{\frac{\alpha^2x^2}{2}}\frac{d^n}{d(\frac{\alpha x}{\sqrt{2}})^n}e^{-\frac{\alpha^2x^2}{2}}\right)e^{-\frac{m\omega}{2\hbar}x^2} \quad ?$$

(4) $$x_{mn} = \langle m|\hat{x}|n\rangle = \sqrt{\frac{\hbar}{2m\omega}}\langle m|(\hat{a}+\hat{a}^\dagger)|n\rangle = \sqrt{\frac{\hbar}{2m\omega}}\left(\sqrt{n}\,\delta_{m,n-1}+\sqrt{n+1}\,\delta_{m,n+1}\right)$$

$$p_{m,n} = \langle m|\hat{p}|n\rangle = \frac{1}{i}\sqrt{\frac{m\hbar\omega}{2}}\langle m|(\hat{a}-\hat{a}^\dagger)|n\rangle = \frac{1}{i}\sqrt{\frac{m\hbar\omega}{2}}\left(\sqrt{n}\,\delta_{m,n-1}-\sqrt{n+1}\,\delta_{m,n+1}\right)$$

$$\langle m|\frac{\hat{p}^2}{2m}|n\rangle = \frac{\hbar\omega}{4}\langle m|\hat{a}\hat{a}^\dagger+\hat{a}^\dagger\hat{a}|n\rangle = \frac{\hbar\omega}{2}\left(n+\frac{1}{2}\right)$$

$$\langle m|\frac{m}{2}\omega^2\hat{x}^2|n\rangle = \frac{\hbar\omega}{4}\langle m|(\hat{a}+\hat{a}^\dagger)^2|n\rangle = \frac{\hbar\omega}{2}\left(n+\frac{1}{2}\right)$$

$$\Delta x = \sqrt{\frac{\hbar}{m\omega}\left(n+\tfrac{1}{2}\right)},\quad \Delta p = \sqrt{m\hbar\omega\left(n+\tfrac{1}{2}\right)} \qquad \Delta x\,\Delta p = \hbar\left(n+\tfrac{1}{2}\right)$$

4.3

(1) $|a\rangle = \sum a_n |n\rangle$, $\hat{a}|a\rangle = a|a\rangle$

$$\hat{a}|a\rangle = \sum_n a_n \sqrt{n}\,|n-1\rangle = \sum_n a_{n+1}\sqrt{n+1}\,|n\rangle = \sum_n a_n\, a\,|n\rangle$$

$$a_{n+1} = \frac{a}{\sqrt{n+1}}\,a_n \quad \to \quad a_n = c\,\frac{a^n}{\sqrt{n!}}$$

$$\langle a|a\rangle = \sum_n |a_n|^2 = c^2 e^{|a|^2} \qquad \therefore \quad |a\rangle = e^{-\frac{|a|^2}{2}} \sum_n \frac{a^n}{\sqrt{n!}}|n\rangle$$

(2) $$\bar{x} = \langle a|\hat{x}|a\rangle = \sqrt{\frac{\hbar}{2m\omega}}\,\langle a|(\hat{a}+\hat{a}^\dagger)|a\rangle = (a+a^*)\sqrt{\frac{\hbar}{2m\omega}}$$

$$\bar{p} = \langle a|\hat{p}|a\rangle = \frac{1}{i}\sqrt{\frac{m\hbar\omega}{2}}\,\langle a|(\hat{a}-\hat{a}^\dagger)|a\rangle = \frac{a-a^*}{i}\sqrt{\frac{m\hbar\omega}{2}}$$

$[\hat{a}, \hat{a}^\dagger] = 1$

$$(\Delta x)^2 = \langle a|(\hat{x}-\bar{x})^2|a\rangle = \langle a|\hat{x}^2|a\rangle - \bar{x}^2 = \frac{\hbar}{2m\omega}\left((a+a^*)^2+1\right) - \bar{x}^2 = \frac{\hbar}{2m\omega}$$

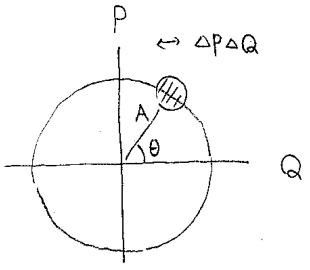
$$(\Delta p)^2 = \langle a|(\hat{p}-\bar{p}^2)|a\rangle = \langle a|\hat{p}^2|a\rangle - \bar{p}^2 = -\frac{m\hbar\omega}{2}\left((a-a^*)^2-1\right) - \bar{p}^2 = \frac{m\hbar\omega}{2}$$

(3) $$\langle a|a'\rangle = e^{-\frac{|a^*|^2+|a'|^2}{2}} \sum_{n,m} \langle m|\frac{a^{*m}}{\sqrt{m!}}\cdot\frac{a'^n}{\sqrt{n!}}|n\rangle = e^{-\frac{|a^*|^2+|a'|^2}{2}} \sum_n \frac{a^{*n}a'^n}{n!}$$

$$= e^{-\frac{|a^*|^2+|a'|^2}{2}}\, e^{a^*a'} = e^{-\frac{|a-a'|^2}{2}}$$

(4) $\hat{a}' = \hat{a} - a$ a: 電場による shift

$\hat{a}'|a\rangle = 0$: grand-state



$\Delta P \Delta Q \sim \Delta A \cdot A \cdot \Delta\theta$

$\simeq \Delta(A^2)\,\Delta\theta$

4.4

(1) $\hat{a}|\alpha\rangle = \sqrt{\frac{m\omega}{2\hbar}}\alpha|\alpha\rangle \rightarrow \left(\hat{x} - \frac{i\hat{p}}{m\omega}\right)|\alpha\rangle = \alpha|\alpha\rangle$

$\left(x + \frac{\hbar}{m\omega}\frac{d}{dx}\right)\psi_\alpha(x) = \alpha\cdot\psi_\alpha(x) \rightarrow \psi_\alpha(x) = \left(\frac{m\omega}{\hbar\pi}\right)^{\frac{1}{4}} e^{-\frac{m\omega}{2\hbar}(x-\alpha)^2}$

(2) $|\alpha\rangle = e^{-\frac{1}{2}\cdot\frac{m\omega}{2\hbar}|\alpha|^2}\sum_n \frac{1}{\sqrt{n!}}\left(\sqrt{\frac{m\omega}{2\hbar}}\alpha\right)^n|n\rangle$

$|\langle n|\alpha\rangle|^2 = \left|e^{-\frac{1}{2}\cdot\frac{m\omega}{2\hbar}|\alpha|^2}\frac{1}{\sqrt{n!}}\left(\sqrt{\frac{m\omega}{2\hbar}}\alpha\right)^n\right|^2 = e^{-\frac{m\omega}{2\hbar}|\alpha|^2}\frac{\left(\frac{m\omega}{2\hbar}|\alpha|^2\right)^n}{n!}$: poisson 分布

$\bar{n} = \langle\alpha|\hat{n}|\alpha\rangle = \langle\alpha|\hat{a}^+\hat{a}|\alpha\rangle = \frac{m\omega}{2\hbar}|\alpha|^2$

(3) $\psi(x,t) = \langle x|e^{-iHt/\hbar}|\alpha\rangle = e^{i\omega\left(\frac{\hbar}{2m\omega}\frac{d^2}{dx^2} - \frac{m\omega}{2\hbar}x^2\right)}\psi_\alpha(x)$

$\langle\alpha(t)|\hat{x}|\alpha(t)\rangle = \langle\alpha|e^{i\hat{H}t/\hbar}\hat{x}\,e^{-i\hat{H}t/\hbar}|\alpha\rangle$

$= \sqrt{\frac{\hbar}{2m\omega}}\langle\alpha|e^{i\omega t\hat{a}^+\hat{a}}(\hat{a}^+ + \hat{a})e^{-i\omega t\hat{a}^+\hat{a}}|\alpha\rangle$

$=$

4.5

$$H = \frac{1}{2m}\left(\hat{P}_x^2 + \hat{P}_y^2 + \hat{P}_z^2\right) + \frac{m}{2}\left(\omega^2\hat{x}^2 + \omega^2\hat{y}^2 + \omega'^2\hat{z}^2\right)$$

$$= \hbar\omega\left(\hat{a}_x^\dagger\hat{a}_x + \frac{1}{2}\right) + \hbar\omega\left(\hat{a}_y^\dagger\hat{a}_y + \frac{1}{2}\right) + \hbar\omega'\left(\hat{a}_z^\dagger\hat{a}_z + \frac{1}{2}\right)$$

(1) $E = \hbar\omega(n_x + n_y + 1) + \hbar\omega'\left(n_z + \frac{1}{2}\right)$

$\Psi_N(x,y,z) = \psi_{n_x,\omega}(x)\,\psi_{n_y,\omega}(y)\,\psi_{n_z,\omega'}(z)$ $\quad N = n_x + n_y + n_z$

$$\left(\psi_{n,\omega}(x) = \sqrt{\frac{m\omega}{n!\,2^n\hbar\sqrt{\pi}}}\,H_n\left(\sqrt{\frac{m\omega}{\hbar}}\,x\right)e^{-\frac{m\omega}{2\hbar}x^2}\right)$$

(2) $\omega' < \omega$ → $(0,0,1)$

$\omega' = \omega$ → $(1,0,0),(0,1,0),(0,0,1)$

$\omega' > \omega$ → $(1,0,0),(0,1,0)$

$\omega' = \omega$ 일 때 $\quad {}_{N+2}C_2 = \frac{(N+2)(N+1)}{2}$

4.6 fan der weels Interaction

(1) $V = -\frac{3}{R^5}(\vec{P}_1\cdot\vec{r})(\vec{P}_2\cdot\vec{r}) + \frac{1}{R^3}(\vec{P}_1\cdot\vec{P}_2) = -\frac{2e^2x_1x_2}{R^3}$ $\quad P_1 = ex_1$, $P_2 = ex_2$

(2) $\hat{H} = \frac{1}{2m}(\hat{P}_1^2+\hat{P}_2^2) + \frac{m\omega^2}{2}\left(\hat{x}_1^2 - \frac{4e^2}{m\omega^2R^3}\hat{x}_1\hat{x}_2 + \hat{x}_2^2\right)$

$\quad\quad\quad

5.1

(1) $L_k = \frac{1}{\hbar}(r\times p)_k = \frac{1}{\hbar} r_i p_j \epsilon_{ijk}$ $\qquad [x, p] = i\hbar$

$$[L_k, r_\ell] = \frac{\epsilon_{ijk}}{\hbar}\left\{ r_i \underbrace{[p_j, r_\ell]}_{-i\hbar\delta_{j\ell}} + \underbrace{[r_i, r_\ell]}_{0} p_j \right\} = -i r_i \epsilon_{i\ell k}$$

$$[L_k, p_\ell] = \frac{\epsilon_{ijk}}{\hbar}\left\{ r_i \underbrace{[p_j, p_\ell]}_{0} + \underbrace{[r_i, p_\ell]}_{i\hbar\delta_{i\ell}} p_j \right\} = i p_j \epsilon_{\ell jk}$$

$$[L_k, L_\ell] = \frac{\epsilon_{ijk}}{\hbar}\left\{ r_i \underbrace{[p_j, L_\ell]}_{-ip_m\epsilon_{jm\ell}} + \underbrace{[r_i, L_\ell]}_{ir_n\epsilon_{ni\ell}} p_j \right\} = \frac{i}{\hbar}\left(-r_i p_m \underbrace{\epsilon_{ijk}\epsilon_{jm\ell}}_{\epsilon_{kij}\epsilon_{mlj}} + r_n p_j \underbrace{\epsilon_{ijk}\epsilon_{ni\ell}}_{\epsilon_{jki}\epsilon_{\ell ni}} \right)$$

$$= -\frac{i}{\hbar} r_i p_m (\delta_{km}\delta_{i\ell} - \delta_{k\ell}\delta_{im}) + \frac{i}{\hbar} r_n p_j (\delta_{j\ell}\delta_{kn} - \delta_{jn}\delta_{k\ell})$$

$$= -\frac{i}{\hbar} r_\ell p_k + \frac{i}{\hbar} r_i p_i \delta_{k\ell} + \frac{i}{\hbar} r_k p_\ell - \frac{i}{\hbar} r_j p_j \delta_{k\ell}$$

$$= \frac{i}{\hbar}(r_k p_\ell - r_\ell p_k) = \frac{i}{\hbar} r_s p_t (\delta_{ks}\delta_{\ell t} - \delta_{kt}\delta_{s\ell}) = i\epsilon_{k\ell m}\frac{1}{\hbar} r_s p_t \epsilon_{stm} = i\epsilon_{k\ell m} L_m \;//$$

$$[L_\pm, L_z] = [L_x, L_z] \pm i[L_y, L_z] = -iL_y \mp L_x = \mp L_\pm \;//$$

$$[L^2, L_z] = [L_x^2, L_z] + [L_y^2, L_z] = L_x[L_x, L_z] + [L_x, L_z]L_x + L_y[L_y, L_z] + [L_y, L_z]L_y$$

$$= -iL_xL_y - iL_yL_x + iL_yL_x + iL_xL_y = 0 \;//$$

$$[L^2, L_\pm] = [L^2, L_x] \pm i[L^2, L_y] = 0 \;//$$

(2) $r^2 = x^2+y^2+z^2$, $\tan\theta = \frac{\sqrt{x^2+y^2}}{z}$, $\tan\phi = \frac{y}{x}$

$$L_x = \frac{1}{i}\left\{ -\sin\phi\frac{\partial}{\partial\theta} - \cot\theta\cos\phi\frac{\partial}{\partial\phi} \right\}$$

$$L_y = \frac{1}{i}\left\{ \cos\phi\frac{\partial}{\partial\theta} - \cot\theta\sin\phi\frac{\partial}{\partial\phi} \right\}$$

$$L_z = \frac{1}{i}\frac{\partial}{\partial\phi}$$

$$L_\pm = e^{\pm i\phi}\left\{ \pm\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\phi} \right\}$$

$$L^2 = -\left\{ \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2} \right\}$$

5.2

(1) $\Phi_m(\phi) = \frac{1}{\sqrt{2\pi}} e^{im\phi}$, $\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta \frac{d\Theta}{d\theta}\right) + \left(-\frac{m^2}{\sin^2\theta} + l(l+1)\right)\Theta = 0$

(2) $[L^2, L_\pm] = 0 \rightarrow L^2(L_\pm Y_{lm}) = L_\pm L^2 Y_{lm} = l(l+1)(L_\pm Y_{lm})$

$[L_z, L_\pm] = \pm L_\pm \rightarrow L_z(L_\pm Y_{lm}) = (L_\pm L_z \pm L_\pm) Y_{lm} = (m \pm 1)(L_\pm Y_{lm})$

(3) $L_+ Y_{lm} = -i\hbar e^{i\phi}\left[i\frac{\partial}{\partial\theta} - \cot\theta \frac{\partial}{\partial\phi}\right] e^{im\phi}\Theta(\theta) = 0 \rightarrow \Theta = c(\sin\theta)^m$

$L^2 Y_{lm} = l(l+1) Y_{lm} \rightarrow \{m(m+1) - l(l+1)\}\sin^m\theta = 0$ $\quad m_{Max} = l$

($L^2 - L_z^2 = L_x^2 + L_y^2 \geq 0 \rightarrow l(l+1) - m^2 \geq 0$)

(4) $Y_{ll} = c_l e^{il\phi} \sin^l\theta$

$Y_{l,l-1} = L_- Y_{ll} = c_l e^{i\phi}\left(-\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\phi}\right) e^{il\phi}\sin^l\theta$

$=$

$$Y_{lm} = \frac{(-1)^l}{2^l l!}\sqrt{\frac{(2l+1)(l+m)!}{4\pi(l-m)!}}\, e^{im\phi}\frac{1}{\sin^m\theta}\cdot\frac{d^{l-m}}{d(\cos\theta)^{l-m}}(\sin\theta)^{2l}$$

(5) $Y_{1,0} = \sqrt{\frac{3}{4\pi}}\cos\theta$

$Y_{1,\pm1} = \mp\sqrt{\frac{3}{8\pi}}\sin\theta \cdot e^{\pm i\phi}$

$Y_{2,0} = \sqrt{\frac{5}{16\pi}}(3\cos^2\theta - 1)$

$Y_{2,\pm1} = \mp\sqrt{\frac{15}{8\pi}}\sin\theta\cdot\cos\theta\, e^{\pm i\phi}$

$Y_{2,\pm2} = \sqrt{\frac{15}{32\pi}}\sin^2\theta\, e^{\pm 2i\phi}$

5.3

(1) $J_\pm |j,m\rangle$ のノルムは

$$\langle jm | J_\mp J_\pm | jm \rangle = \langle jm | \left(J^2 - J_z^2 \pm i[J_x, J_y] \right) | jm \rangle = \left\{ \lambda_j - m(m\pm 1) \right\} \langle jm | jm \rangle \geq 0$$

$$\lambda_j \geq m(m\pm 1)$$

(2) $J_+ | j\, m_{Max} \rangle = 0$

$$J_- J_+ | j\, m_{Max} \rangle = \left(J^2 - J_z^2 + i[J_x, J_y] \right) | j\, m_{Max} \rangle = 0$$

$$\left.\begin{aligned} \lambda_j - m_{Max}(m_{Max}+1) &= 0 \\ \lambda_j - m_{Min}(m_{Min}-1) &= 0 \end{aligned}\right\} \quad (m_{Max} + m_{Min})(m_{Max} - m_{Min} + 1) = 0$$

$(m_{Max} + m_{Min}) = 0$, $(m_{Max} - m_{Min} + 1) \neq 0$

$$m_{Max} = m_{Min} + n \quad \longrightarrow \quad m_{Max} = \frac{n}{2} \ (\text{整数 or 半整数})$$

$$\lambda_j = j(j+1)$$

(3) $\langle jm | J_- J_+ | jm \rangle = (j-m)(j+m+1)$

$$J_\pm | j\, m \rangle = \sqrt{(j \mp m)(j \pm m + 1)}\, | j\, m \pm 1 \rangle$$

$$\langle j'm' | J_\pm | jm \rangle = \sqrt{(j \mp m)(j \pm m + 1)}\, \delta_{jj'} \delta_{m', m\pm 1}$$

$$\langle j'm' | J_z | jm \rangle = m\, \delta_{jj'} \delta_{mm'}$$

$j = \frac{1}{2}$: $J_z = \begin{pmatrix} \frac{1}{2} & \\ & -\frac{1}{2} \end{pmatrix}$ $J_+ = \begin{pmatrix} & 1 \\ & \end{pmatrix}$ $J_- = \begin{pmatrix} & \\ 1 & \end{pmatrix}$ $J_x = \begin{pmatrix} & \frac{1}{2} \\ \frac{1}{2} & \end{pmatrix}$ $J_y = \begin{pmatrix} & -\frac{i}{2} \\ \frac{i}{2} & \end{pmatrix}$

5.4

(1) $\Delta = \frac{1}{r^2}\partial_r(r^2\partial_r) - \frac{\hat{L}^2}{r^2}$ $\qquad \Delta\psi + \frac{2m}{\hbar^2}(E-V)\psi = 0$

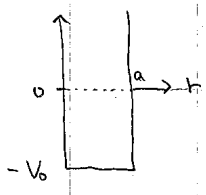
$\psi = R(r)Y(\theta,\varphi)$ $\qquad L^2 Y = l(l+1)Y$

$R(r) = \frac{u(r)}{r}$ $\qquad \frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \frac{2mr^2}{\hbar^2}(E-V)R = l(l+1)R$

$$\frac{d^2u}{dr^2} + \left[\underbrace{\frac{2m}{\hbar^2}(E-V)}_{k^2} - \frac{l(l+1)}{r^2}\right]u = 0$$

(2) $\underline{l=0}$ $\qquad u_0'' + k^2 u_0 = 0$

$R_0 = \frac{a}{2\pi}\frac{\sin kr}{r}$



$\underline{l=1}$ $\qquad R_1'' + \frac{2}{r}R_1' + (k^2 - \frac{2}{r^2})R_1 = 0$

$R_0'' + \frac{2}{r}R_0' + k^2 R_0 = 0 \rightarrow R_0''' + \frac{2}{r}R_0'' + (k^2 - \frac{2}{r^2})R_0' = 0$

$R_1 \propto R_0' \propto \frac{\sin kr}{r^2} - \frac{\cos kr}{r}$

(3)

6.1

(1) $l = 1,\ m = 0, \pm 1$ $\quad 2l+1 = 3$ $\quad 3^2$ 通り

(2) $L^2 = (L_1 + L_2)^2$ $\quad L^2 = L(L+1) = 0, 2, 6$

(3)

$L = 2$ ← 出発点

$|L=2, M=2\rangle = |11\rangle$

$|L=2, M=1\rangle = \frac{1}{2} L_- |L=2, M=2\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle)$

$|L=2, M=0\rangle = \frac{1}{\sqrt{6}} L_- |L=2, M=1\rangle = \frac{1}{\sqrt{6}} (|1-1\rangle + 2|00\rangle + |1-1\rangle)$

$|L=2, M=-1\rangle = \frac{1}{\sqrt{2}} (|-10\rangle + |0-1\rangle)$

$|L=2, M=-2\rangle = |-1-1\rangle$

$L = 1$

$|L=1, M=1\rangle = A|10\rangle + B|01\rangle \perp |L=2, M=1\rangle$

$= \frac{1}{\sqrt{2}} (|10\rangle - |01\rangle)$

$|L=1, M=0\rangle = \frac{1}{\sqrt{2}} (|1-1\rangle - |-11\rangle)$

$|L=1, M=-1\rangle = \frac{1}{\sqrt{2}} (|0-1\rangle - |-10\rangle)$

$L = 0$

$|L=0\ M=0\rangle = A|1-1\rangle + B|00\rangle + C|-11\rangle \perp |L=2\ M=0\rangle, |L=1, M=0\rangle$

$= \frac{1}{\sqrt{3}} (|1-1\rangle - |00\rangle + |-11\rangle)$

(4)

6.2

(1) $(J_-)^{j-m} |j_1 j_2 ; j j\rangle = (J_-)^{j-m-1} \sqrt{2j \cdot 1}\, |j_1 j_2 ; j\, j-1\rangle$

$= \sqrt{(2j)(2j-1)\cdots((j+m+1)\cdot 1\cdot 2\cdot\cdots(j-m)}\ |j_1 j_2 ; j\, m\rangle$

$\frac{(2j)!\,(j-m)!}{(j+m)!}$

(2) $|j_1 j_2 ; J\, J\rangle = |j_1 j_2 ; j j\rangle$ $\qquad J = j_1 + j_2 = 2j$

$|j_1 j_2 ; J\, J-1\rangle = \frac{1}{\sqrt{2J}} J_- |j_1 j_2 ; j j\rangle = \frac{1}{\sqrt{2}} \left(|j_1 j_2 ; j\, j-1\rangle + |j_1 j_2 ; j\, j-1\rangle \right)$

$|j_1 j_2 ; J\, J-2\rangle = \frac{1}{\sqrt{2J(2J-1)2}} J_-^2 |j_1 j_2 ; j j\rangle =$

$= \frac{1}{2\sqrt{2J-1}} J_- \left(|j_1 j_2 ; j-1\, j\rangle + |j_1 j_2 ; j\, j-1\rangle \right)$

$= \frac{1}{2\sqrt{2J-1}} \left(\sqrt{(2j-1)2}\, |j_1 j_2 ; j\, j-2\rangle + 2\sqrt{2j}\, |j_1 j_2 ; j-1\, j-1\rangle + \sqrt{(2j-1)2}\, |j_1 j_2 ; j\, j-2\rangle \right)$

$= \sqrt{\frac{2j-1}{2(4j-1)}}\, |j_1 j_2 ; j\, j-2\rangle + \sqrt{\frac{2j}{4j-1}}\, |j_1 j_2 ; j-1\, j-1\rangle + \sqrt{\frac{2j-1}{2(4j-1)}}\, |j_1 j_2 ; j\, j-2\rangle$

(3) $(j_1 = \frac{1}{2} \quad j_2 = \frac{1}{2})$

$\underline{J=1}$ $\quad |J=1\ M=1\rangle = |++\rangle$

$|J=1\ M=0\rangle = \frac{1}{\sqrt{2}} (|+-\rangle + |-+\rangle)$

$|J=1\ M=-1\rangle = |--\rangle$

$\underline{J=0}$ $\quad |J=0\ M=0\rangle = \frac{1}{\sqrt{2}} (|+-\rangle - |-+\rangle)$

$(j_1 = 1 \quad j_2 = \frac{1}{2})$

$\underline{J=\frac{3}{2}}$ $\quad |J=\frac{3}{2}\ M=\frac{3}{2}\rangle = |1\ \tfrac{1}{2}\rangle$

$|J=\frac{3}{2}\ M=\frac{1}{2}\rangle = \frac{1}{\sqrt{3}} J_- |1\ \tfrac{1}{2}\rangle = \sqrt{\frac{2}{3}}\, |0\ \tfrac{1}{2}\rangle + \frac{1}{\sqrt{3}}\, |1\ -\tfrac{1}{2}\rangle$

$|J=\frac{3}{2}\ M=-\frac{1}{2}\rangle = \frac{1}{2} J_- \left(\sqrt{\frac{2}{3}}\, |0\ \tfrac{1}{2}\rangle + \frac{1}{\sqrt{3}} |1\ -\tfrac{1}{2}\rangle \right) = \frac{1}{\sqrt{6}} \left(\sqrt{2}\, |-1\ \tfrac{1}{2}\rangle + |0\ -\tfrac{1}{2}\rangle \right) + \frac{1}{2\sqrt{3}} \sqrt{2}\, |0\ -\tfrac{1}{2}\rangle$

$= \frac{1}{\sqrt{3}} |-1\ \tfrac{1}{2}\rangle + \sqrt{\frac{2}{3}}\, |0\ -\tfrac{1}{2}\rangle$

$|J=\frac{3}{2}\ M=-\frac{3}{2}\rangle = \frac{1}{\sqrt{3}} J_- \left(\frac{1}{\sqrt{3}} |-1\ \tfrac{1}{2}\rangle + \frac{2}{\sqrt{6}} |0\ -\tfrac{1}{2}\rangle \right) = \frac{1}{3} |-1\ -\tfrac{1}{2}\rangle + \frac{2}{3} |-1\ -\tfrac{1}{2}\rangle = |-1\ -\tfrac{1}{2}\rangle$

$\underline{J=\frac{1}{2}}$ $\quad |J=\frac{1}{2}\ M=\frac{1}{2}\rangle = \frac{1}{\sqrt{3}} |0\ \tfrac{1}{2}\rangle - \sqrt{\frac{2}{3}}\, |1\ -\tfrac{1}{2}\rangle$

$|J=\frac{1}{2}\ M=-\frac{1}{2}\rangle = J_- \left(\frac{1}{\sqrt{3}} |0\ \tfrac{1}{2}\rangle - \sqrt{\frac{2}{3}}\, |1\ -\tfrac{1}{2}\rangle \right) = \sqrt{\frac{2}{3}}\, |-1\ \tfrac{1}{2}\rangle + \frac{1}{\sqrt{3}} |0\ -\tfrac{1}{2}\rangle - \frac{2}{\sqrt{3}} |0\ -\tfrac{1}{2}\rangle$

$= \sqrt{\frac{2}{3}}\, |-1\ \tfrac{1}{2}\rangle - \frac{1}{\sqrt{3}} |0\ -\tfrac{1}{2}\rangle$

6.3

$|L-S| \leq J \leq L+S$

(1) $L=2, S=\frac{1}{2}$ $(2L+1)(2S+1)=10$ $J=\frac{5}{2},\frac{3}{2}$ $(2\cdot\frac{5}{2}+1)+(2\cdot\frac{3}{2}+1)=10$

(2) $\hat{H}=\hat{H}_0+\xi\hat{L}\cdot\hat{S}$

$\hat{H}_0$ の量子数 1° $\hat{L}^2, \hat{L}_z, \hat{S}^2, \hat{S}_z$

2° $\hat{L}^2\ \hat{S}^2, \hat{J}^2, \hat{J}_z$ ← $\hat{L}\cdot\hat{S}=\frac{1}{2}(\hat{J}^2-\hat{L}^2-\hat{S}^2)$ と可換

$\hat{L}\cdot\hat{S}=\frac{1}{2}(J(J+1)-L(L+1)-S(S+1))$

$|L-S| \leq J \leq L+S$ $J=L+S, L-S$ $\Delta J=1$

$\Delta E=\frac{\xi}{2}((J+1)(J+2)-J(J+1))=\xi(J+1)$

(3) $\hat{H}_0=\frac{\hat{p}^2}{2m}+V(\hat{r})=-\frac{\hbar^2}{2m}\frac{1}{r^2}\partial_r(r^2\partial_r)+\frac{\hbar^2}{2mr^2}\hat{L}^2+V(\hat{r})$

$\hat{H}\psi(\vec{r})=\left[-\frac{\hbar^2}{2m}\frac{1}{r^2}\partial_r(r^2\partial_r)+V(r)+\frac{l(l+1)\hbar^2}{2mr^2}+\frac{\xi}{2}(j(j+1)-l(l+1)-\underbrace{s(s+1)}_{\frac{3}{4}})\right]\psi(\vec{r})$

7.1

(1) 並進 $\hat{U}(\vec{\delta})\psi(x_k) = \psi(x_k+\delta a_k) = (1+\delta a_k \cdot \partial_k)\psi(x_k)$

$\partial_k = \frac{i}{\hbar}\hat{P}_k$

回転 $\hat{U}(\vec{\delta})\psi(x_k) = \psi(x_k+\delta\omega_{ki}x_i) = (1+\delta\omega_{ki}x_i\partial_k)\psi(x_k)$

$\delta\omega_i = \epsilon_{ijk}\delta\omega_{jk}$

$\frac{i}{\hbar}\delta\vec{\omega}\cdot\vec{L} = \frac{i}{\hbar}\delta\vec{\omega}\cdot(\vec{x}\times\vec{P}) = \epsilon_{klm}\delta\omega_{lm}\cdot\epsilon_{ijk}x_i\partial_j = (\delta_{il}\delta_{jm}-\delta_{im}\delta_{jl})\delta\omega_{lm}x_i\partial_j$

$= \delta\omega_{ij}x_i\partial_j - \delta\omega_{ji}x_i\partial_j$

(2) $\hat{U}(\vec{\delta})\hat{x}_i\hat{U}^+(\vec{\delta}) = (1+\frac{i}{\hbar}\vec{\delta}\cdot\vec{O})\hat{x}_i(1-\frac{i}{\hbar}\vec{\delta}\cdot\vec{O}^+) = \hat{x}_i + \frac{i}{\hbar}\vec{\delta}\cdot[\vec{O},\hat{x}_i]$ （$\vec{O}=0$）

並進 $= \hat{x}_i + \frac{i}{\hbar}\vec{\delta}\cdot[\hat{x}_i,\vec{P}] = \hat{x}_i + \delta_i$

回転 $= \hat{x}_i - \frac{i}{\hbar}\vec{\delta}\cdot[\hat{x}_i,\vec{L}] = \hat{x}_i + \delta\omega_{ij}\hat{x}_j$

$\delta_j\cdot\epsilon_{klm}[x_i,x_lP_m] = i\hbar\epsilon_{jli}\delta_j x_l$

$\hat{U}(\vec{\delta})\hat{P}_i\hat{U}^+(\vec{\delta}) =$ 並進 $= P_i$

回転 $= P_i - \frac{i}{\hbar}\vec{\delta}\cdot[\hat{P}_i,\vec{L}]$

(3) $i\hbar\frac{\partial}{\partial t}\psi'(x) = i\hbar\frac{\partial}{\partial t}\hat{U}(\vec{\delta})\psi(x) = \hat{U}(\vec{\delta})i\hbar\frac{\partial}{\partial t}\psi(x)$

$= \hat{U}(\vec{\delta})\hat{H}\psi(x) = \hat{H}\hat{U}(\vec{\delta})\psi(x)$ ← $[\vec{\delta},\hat{H}]=0$

$= \hat{H}\psi'(x)$

回転 $\vec{x} = x^i\vec{e}_i$ $\quad \delta\vec{x} = \delta x^i\cdot\vec{e}_i + x^i\delta\vec{e}_i$

$\delta x^k = \delta x^i e_i^k + x^i\delta e_i^k$

↑ i軸のk成分

$= \delta x^i e_i^k + \omega_{ij}x^i e_j^k$

$e_i^k e_j^k = \delta_{ij} \rightarrow \delta e_i^k e_j^k + e_i^k\delta e_j^k = 0$

$\delta e_i^k = \omega_{ij}e_j^k$

$= -\omega_{ji}$

$\omega_k = \epsilon_{ijk}\omega_{ij}$

$(\vec{\omega}\times\vec{x})_k = \epsilon_{ijk}\omega_i x^j$

$= \epsilon_{ijk}\epsilon_{lmi}\omega_{lm}x^j$

$= (\delta_{jl}\delta_{km}-\delta_{jm}\delta_{kl})\omega_{lm}x^j$

$= \omega_{jk}x^j - \omega_{kj}x^j$

7.2

(1) $\hat{H} = \frac{\hat{p}^2}{2m}$ $\quad \hat{O} = \hat{P}, \hat{L}, \hat{\pi}$ パリティ変換

$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{r})$ $\quad \hat{O} = \hat{L}, \hat{\pi}$

$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2(\hat{x}^2+\hat{y}^2) + \frac{1}{2}m\omega'^2\hat{z}^2$ $\quad \hat{O} = \hat{L}_z, \hat{\pi}$

(2) (a) $\hat{H}\psi' = \hat{H}\hat{U}\psi = \hat{U}\hat{H}\psi = E\hat{U}\psi = E\psi'$ より縮退

(b) (i) 並進 → 平面波

(ii) 回転 → 中心力場の解

(iii) パリティ → 偶関数

7.3

(1) $H = \frac{1}{2m}(p_x^2+p_y^2+p_z^2) + \frac{m\omega^2}{2}(x^2+y^2+z^2) = \hbar\omega\left(n_x+n_y+n_z+\frac{3}{2}\right)$

$|000\rangle = \left(\frac{m^2\omega^2}{\pi\hbar^2}\right)^{3/4} e^{-\frac{m\omega}{2\hbar}(x^2+y^2+z^2)}$

$|100\rangle = \hat{a}_x^+ |000\rangle$

$\hat{U}(\Theta) = e^{-\frac{i}{\hbar}\Theta\hat{L}_z}$ $\quad \hat{L}_z = \hat{x}\hat{P}_y - \hat{y}\hat{P}_x = \frac{\hbar}{i}(\hat{a}_x^+\hat{a}_y - \hat{a}_x\hat{a}_y^+)$

$\hat{L}_z^n |100\rangle = \begin{cases} i|010\rangle & (\text{odd}) \\ |100\rangle & (\text{even}) \end{cases}$ $\quad \hat{U}(\Theta)|100\rangle = \cos\Theta|100\rangle + \sin\Theta|010\rangle$

$\hat{L}_z^n |010\rangle = \begin{cases} -i|100\rangle & (\text{odd}) \\ |010\rangle & (\text{even}) \end{cases}$ $\quad \hat{U}(\Theta)|010\rangle = -\sin\Theta|100\rangle + \cos\Theta|010\rangle$

$\hat{L}_z^n |001\rangle = \begin{cases} |001\rangle & (n=0) \\ 0 & (n>0) \end{cases}$ $\quad \hat{U}(\Theta)|001\rangle = |001\rangle$

(2) $Y_{lm}(\pi-\theta, \varphi+\pi) = (-1)^l Y_{lm}(\theta,\varphi)$ $\quad \hat{\pi}$ と $\hat{L}$ は同時固有 fn がある

7.4